

# HW2 - STAT 709

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## Notation and conventions

All random variables are defined on a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ , and  $\mathbb{E}$  denotes expectation with respect to  $\mathbb{P}$ . For  $1 \leq p < \infty$ , write

$$\|X\|_p = (\mathbb{E}|X|^p)^{1/p}, \quad \|X\|_\infty = \inf\{M \geq 0 : \mathbb{P}(|X| \leq M) = 1\}.$$

If  $F$  is a cdf, then  $dF$  denotes the probability measure induced by  $F$ , so integrals with respect to  $dF$  are Lebesgue–Stieltjes integrals. We use  $s \vee a = \max\{s, a\}$ , and  $G \sim N(0, 1)$  means that  $G$  is a standard normal random variable. Statements about equality or boundedness of random variables are understood almost surely when appropriate.

## Problem 1. Ex.35. (a),(b)

Let  $\{a_n\}_{n \geq 1}$  be a sequence of positive numbers satisfying

$$\sum_{n=1}^{\infty} a_n = 1,$$

and let  $\{P_n\}_{n \geq 1}$  be a sequence of probability measures on a common measurable space. Define

$$P = \sum_{n=1}^{\infty} a_n P_n.$$

- (a) Show that  $P$  is a probability measure.
- (b) For a measure  $\nu$ , show that  $P_n \ll \nu$  for all  $n$  if and only if  $P \ll \nu$ . Moreover, when  $P \ll \nu$  and  $\nu$  is  $\sigma$ -finite, show that

$$\frac{dP}{d\nu} = \sum_{n=1}^{\infty} a_n \frac{dP_n}{d\nu}.$$

**Sol.**

- (a) We know that  $P$  is a measure from the Ex 1.7. in HW1. Now only need to check that

$$P(\Omega) = \sum_{n=1}^{\infty} a_n P_n(\Omega) = \sum_{n=1}^{\infty} a_n = 1$$

and we are done.

- (b) To see the sufficiency, we know for any measurable set  $A$ ,

$$P(A) \geq a_n P_n(A) \implies P_n(A) \leq P(A)/a_n$$

and hence if  $\nu(A) = 0$ , then  $P_n(A) \leq P(A)/a_n = 0$  for any  $n$ , which means  $P_n \ll \nu$  for any  $n$ . The

necessity is easy to check by noticing if  $\nu(A) = 0$ , then  $P_n(A) = 0$  for any  $n$  and hence

$$P(A) = \sum_{n=1}^{\infty} a_n P_n(A) = \sum_{n=1}^{\infty} a_n \cdot 0 = 0.$$

To sum up, we showed the required equivalence.

If  $P \ll \nu$ , with  $\nu$   $\sigma$ -finite, we know that  $P_n \ll \nu$  by the conclusion above. Denote

$$f = \sum_{n=1}^{\infty} a_n \frac{dP_n}{d\nu}$$

and for any measurable set  $A$ , by the MCT, we know

$$\int_A f d\nu = \sum_{n=1}^{\infty} a_n \int_A \frac{dP_n}{d\nu} d\nu = \sum_{n=1}^{\infty} a_n P_n(A) = P(A)$$

and hence  $f = \frac{dP}{d\nu}$  by the Radon-Nikodym Theorem.

### Problem 2. Ex.40.

Let  $X$  be a random variable having a continuous c.d.f.  $F$ . Show that  $Y = F(X)$  has the uniform distribution  $U(0, 1)$ .

**Sol.**

Define

$$F^{-1}(a) = \sup\{y, F^{-}(y) < a\}$$

and we know  $F(F^{-1}(a)) \geq a$  by the right-continuity. Notice

$$P(Y \geq a) = P(F(X) \geq a) = P(X \geq F^{-1}(a)) = 1 - F^{-}(F^{-1}(a))$$

where

$$F^{-}(F^{-1}(a)) \leq a$$

since  $F^{-}$  is left-continuous. However,  $F^{-} = F$  since  $F$  continuous and hence  $F^{-}(F^{-1}(a)) = a$ . Then we know

$$P(Y \geq a) = 1 - a$$

and we are done.

### Problem 3. Ex.55.

Let  $X$  be a random variable. Show that

(a) if  $EX$  exists, then

$$EX = \int_0^{\infty} P(X > x) dx - \int_{-\infty}^0 P(X \leq x) dx$$

(b) if  $X$  has range  $\mathbb{N}$ , then

$$EX = \sum_{n \geq 1} P(X \geq n)$$

**Sol.**

(a) Notice that by the Fubini's Theorem, we have

$$\begin{aligned}
 EX &= \int_{-\infty}^{\infty} y d\mu(y) \\
 &= \int_{(0,\infty)} y d\mu(y) - \int_{(-\infty,0]} (-y) d\mu(y) \\
 &= \int_{(0,\infty)} \int_{(0,y)} dx d\mu(y) - \int_{(-\infty,0]} \int_{[y,0)} dx d\mu(y) \\
 &= \int_0^{\infty} \int_{(x,\infty)} d\mu(y) dx - \int_{-\infty}^0 \int_{(-\infty,x]} d\mu(y) dx \\
 &= \int_0^{\infty} P(X > x) dx - \int_{-\infty}^0 P(X \leq x) dx
 \end{aligned}$$

where  $\mu$  is the induced measure by  $X$ .

(b) Use the inequality in (a) directly and we have

$$\begin{aligned}
 EX &= \int_0^{\infty} P(X > x) dx - \int_{-\infty}^0 P(X \leq x) dx \\
 &= \sum_{n=0}^{\infty} \int_n^{n+1} P(X \geq n+1) - \int_0^0 P(X = 0) dx \\
 &= \sum_{n \geq 1} P(X \geq n)
 \end{aligned}$$

and we are done.

#### Problem 4. Ex.83.

Let  $X$  be an integrable random variable with a Lebesgue p.d.f.  $f_X$  and let  $Y = g(X)$ , where  $g$  is a function with positive derivative on  $(0, \infty)$  and  $g(x) = g(-x)$ . Find an expression for  $E(X|Y)$  and verify that it is indeed the conditional expectation.

**Sol.**

Let

$$h = \left( g \Big|_{[0,\infty)} \right)^{-1}$$

be the inverse of the restriction of  $g$  to  $[0, \infty)$ . Since  $g$  is even and strictly increasing on  $[0, \infty)$ , we have

$$h(Y) = h(g(X)) = |X| \quad \text{a.s.}$$

For  $u \geq 0$ , define

$$r(u) = \begin{cases} \frac{f_X(u) - f_X(-u)}{f_X(u) + f_X(-u)}, & f_X(u) + f_X(-u) > 0, \\ 0, & f_X(u) + f_X(-u) = 0. \end{cases}$$

and we prove that

$$E(X | Y) = h(Y)r(h(Y)).$$

Denote

$$Z = h(Y)r(h(Y)).$$

Since  $Z$  is a Borel function of  $Y$ , it is  $Y$ -measurable. Moreover,  $|r(u)| \leq 1$ , and hence

$$|Z| \leq h(Y) = |X|.$$

Since  $X$  is integrable,  $Z$  is also integrable.

Let  $\varphi$  be any bounded Borel function. Using the fact that  $g(-u) = g(u)$ , we obtain

$$\begin{aligned} E[\varphi(Y)X] &= \int_{\mathbb{R}} \varphi(g(x))x f_X(x) dx \\ &= \int_0^\infty \varphi(g(u))u f_X(u) du - \int_0^\infty \varphi(g(u))u f_X(-u) du \\ &= \int_0^\infty \varphi(g(u))u (f_X(u) - f_X(-u)) du. \end{aligned}$$

On the other hand,

$$\begin{aligned} E[\varphi(Y)Z] &= \int_{\mathbb{R}} \varphi(g(x))h(g(x))r(h(g(x)))f_X(x) dx \\ &= \int_0^\infty \varphi(g(u))ur(u)(f_X(u) + f_X(-u)) du \\ &= \int_0^\infty \varphi(g(u))u(f_X(u) - f_X(-u)) du. \end{aligned}$$

Therefore,

$$E[\varphi(Y)Z] = E[\varphi(Y)X]$$

for every bounded Borel function  $\varphi$  and we are done.

### Problem 5. Integration by parts

Let  $a < b$  and  $f, g \in C^1([a, b])$ , the space of continuously differentiable functions on  $[a, b]$ . The usual integration-by-parts formula is

$$\int_a^b f'(x)g(x) dx = f(b)g(b) - f(a)g(a) - \int_a^b f(x)g'(x) dx.$$

We can regard  $f'(x) dx$  and  $g'(x) dx$  as defining signed measures on  $[a, b]$ . If the derivatives are non-negative, these are nonnegative measures. We now generalize the formula to a probability measure with cdf  $F$ . Recall that probability measures on  $\mathbb{R}$  correspond one-to-one to cdfs. For  $g \in C^1([a, b])$ , prove that

$$\int_{(a,b]} g(x) dF(x) = F(b)g(b) - F(a)g(a) - \int_a^b g'(x)F(x) dx.$$

**Sol.**

Notice the by the Fubini's theorem we have

$$\begin{aligned}
& \int_a^b g'(x)F(x) dx + \int_{(a,b]} g(x) dF(x) \\
&= \int_a^b g'(x) \int_{(-\infty, x]} dF(s) dx + \int_{(a,b]} g(s) dF(s) \\
&= \int_{(-\infty, b]} \int_{s \vee a}^b g'(x) dx dF(s) + \int_{(a,b]} g(s) dF(s) \\
&= \int_{(-\infty, b]} (g(b) - g(s \vee a)) dF(s) + \int_{(a,b]} g(s) dF(s) \\
&= \int_{(-\infty, a]} (g(b) - g(a)) dF(s) + \int_{(a,b]} (g(b) - g(s)) dF(s) + \int_{(a,b]} g(s) dF(s) \\
&= (g(b) - g(a))F(a) + g(b)(F(b) - F(a)) \\
&= g(b)F(b) - g(a)F(a),
\end{aligned}$$

and we are done.

### Problem 6. Moment bounds

Let  $X$  be a real-valued random variable.

- (1) Suppose  $0 < p < q$  and  $\mathbb{E}|X|^q < \infty$ . Use Hölder's inequality to show that  $\mathbb{E}|X|^p < \infty$  and

$$(\mathbb{E}|X|^p)^{1/p} \leq (\mathbb{E}|X|^q)^{1/q}.$$

- (2) Suppose

$$M = \sup_{p \geq 1} (\mathbb{E}|X|^p)^{1/p} < \infty.$$

Prove that  $\mathbb{P}(|X| \leq M) = 1$ .

- (3) Let  $G \sim N(0, 1)$ . Prove that there is a constant  $c > 0$  such that

$$\lim_{n \rightarrow \infty} \frac{(\mathbb{E}|G|^n)^{1/n}}{\sqrt{n}} = c,$$

and determine  $c$ .

**Sol.**

- (1) By Hölder's inequality, with conjugate exponents  $q/p$  and  $q/(q-p)$ ,

$$\begin{aligned}
\mathbb{E}|X|^p &= \mathbb{E}(|X|^p \cdot 1) \\
&\leq (\mathbb{E}|X|^{p(q/p)})^{p/q} (\mathbb{E}1^{q/(q-p)})^{(q-p)/q} \\
&= (\mathbb{E}|X|^q)^{p/q}.
\end{aligned}$$

Therefore  $\mathbb{E}|X|^p < \infty$ , and taking  $p$ th roots gives

$$(\mathbb{E}|X|^p)^{1/p} \leq (\mathbb{E}|X|^q)^{1/q}.$$

- (2) Fix  $\varepsilon > 0$  and write

$$\alpha = \mathbb{P}(|X| \geq M + \varepsilon).$$

If  $\alpha > 0$ , then for every  $p \geq 1$ ,

$$\begin{aligned} \mathbb{E}|X|^p &\geq (M + \varepsilon)^p \alpha, \\ \implies (\mathbb{E}|X|^p)^{1/p} &\geq (M + \varepsilon) \alpha^{1/p}. \end{aligned}$$

Let  $p \rightarrow \infty$  contradicts the definition of  $M$ , and hence  $\mathbb{P}(|X| \geq M + \varepsilon) = 0$  for every  $\varepsilon > 0$ , and therefore

$$\mathbb{P}(|X| > M) = 0, \quad \mathbb{P}(|X| \leq M) = 1.$$

(3) Let  $m_n = \mathbb{E}|G|^n$ . Integration by parts gives, for  $n \geq 2$ ,

$$\begin{aligned} m_n &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} |x|^n e^{-x^2/2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} |x|^{n-2} x (x e^{-x^2/2}) dx \\ &= (n-1) \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} |x|^{n-2} e^{-x^2/2} dx \\ &= (n-1) m_{n-2}. \end{aligned}$$

In particular,

$$m_{2k} = (2k-1)(2k-3) \cdots 1 = \frac{(2k)!}{2^k k!}.$$

By Stirling's formula,

$$\begin{aligned} \frac{m_{2k}^{1/(2k)}}{\sqrt{2k}} &= \frac{1}{\sqrt{2k}} \left( \frac{(2k)!}{2^k k!} \right)^{1/(2k)} \simeq \frac{1}{\sqrt{2k}} \left( \frac{(2k)!}{\left( \sqrt{2\pi(2k)} \left( \frac{2k}{e} \right)^{2k} \right)} \frac{2^k \left( \sqrt{2\pi k} \left( \frac{k}{e} \right)^k \right)^{1/2k}}{2^k k!} \right)^{1/2k} \frac{\sqrt{2}^{1/2k} \sqrt{2k}}{\sqrt{e}} \\ &\longrightarrow \frac{1}{\sqrt{e}}. \end{aligned}$$

The  $L^p$  norms are increasing, so

$$m_{2k}^{1/(2k)} \leq m_{2k+1}^{1/(2k+1)} \leq m_{2k+2}^{1/(2k+2)}.$$

and since

$$\lim_{k \rightarrow \infty} \frac{\sqrt{2k+1}}{\sqrt{2k}} = \lim_{k \rightarrow \infty} \frac{\sqrt{2k+1}}{\sqrt{2k+2}} = 1,$$

we know both bounds have the same limit  $e^{-1/2}$ . Therefore, we have

$$\lim_{n \rightarrow \infty} \frac{(\mathbb{E}|G|^n)^{1/n}}{\sqrt{n}} = c,$$

with  $c = e^{-1/2}$ .